

Leo Shaposhnik



Overview

I am a theoretical physicist interested in mathematical structures underlying physical theories. My PhD is focussed on the relation between quantum information theory and the structure of the continuum in quantum field theory and quantum gravity. Over the past six months I have transitioned to a strongly agentic research workflow. A primary interest that developed is how to structure the interaction with agents to become reliable research assistants, whose generated artifacts I can check naturally, while maintaining the gain in efficiency.

Current Position

PhD in Theoretical Physics

Freie Universität Berlin

Berlin, Germany

04/2023 – present

- Research at the intersection of quantum information theory, quantum many-body physics and quantum gravity.
- Co-organize a weekly seminar on quantum many-body physics.
- Taught classes on quantum mechanics, gravity and quantum information.
- Work primarily in international collaborations across several continents.
- Extensive use of agentic coding tools to execute research related tasks and automate scientific discovery.

Previous Experience

IT Consultant

d-fine GmbH

Berlin, Germany

11/2021 – 03/2023

- Implemented a production-grade IFRS17 reporting backend for an international reinsurer in a Scrum environment.

Working Student — Mathematical Modelling and Data Analysis

Physikalisch-Technische Bundesanstalt (PTB)

Berlin, Germany

07/2020 – 10/2021

- Numerical and analytical study of roundoff-induced effects in classical chaotic systems.

Working Student

DESY

Zeuthen, Germany

03/2017 – 06/2020

- Design and implementation of optical experiments to measure components for a linear accelerator.
- Analysis of accelerator components for their concentration of radioactive materials using scripts I maintained.

Publications

- **On Infinite Tensor Networks, Complementary Recovery and Type II Factors.** W. Chemissany, E. Gesteau, A. Jahn, D. Murphy, **L. Shaposhnik**. J. Phys. A 58 (2025) 435301.
- **Searching for Emergent Spacetime in Spin Glasses.** D. Saraidaris, **L. Shaposhnik**. JHEP 04 (2026) 032.

Awards & Honors

- Honorable mention for my poster "On Infinite Tensor Networks, Complementary Recovery and Type II Factors". IQQG 2025, Perimeter Institute.

Education

Master's Degree — Physics (Avg. Grade: 1.0 / GPA: 4.0)

Humboldt Universität zu Berlin / Albert Einstein Institute

Berlin, Germany

10/2019 – 02/2022

Thesis on entanglement entropy in curved spacetime, including analytical derivations and numerical validation using tensor network techniques.

Bachelor's Degree — Physics (Avg. Grade: 1.7 / GPA: 3.3)

Humboldt Universität zu Berlin

Berlin, Germany

10/2016 – 09/2019

Thesis on Hopf-algebraic structures underlying perturbative renormalization in quantum field theory.

Languages

- **English** (Fluent)
- **German** (Native)
- **Russian** (Native)
- **Ukrainian** (Beginner)

Talks & Posters

- **Observers and algebras in quantum gravity.** Talk, DPG Frühjahrsfachtagung, 18 Mar 2024.
- **Chaos and emergent algebras.** Talk, AG Eisert group meeting (FU Berlin), 11 Jan. 2025.
- **On Infinite Tensor Networks, Complementary Recovery and Type II Factors.** Invited Talk, Information, Geometry & Physics Seminar (Caltech), 18 Jun. 2025.
- **On Infinite Tensor Networks, Complementary Recovery and Type II Factors.** Poster, IQQG 2025 (Perimeter Institute), 23–27 Jun 2025.
- **On Infinite Tensor Networks, Complementary Recovery and Type II Factors.** Talk, Panorama of Tensor Networks 2025 (Jagiellonian University, Krakow), 05–10 Oct 2025.
- **Searching for emergent spacetime in spin glasses.** Talk, AG Eisert group meeting (FU Berlin), 23–27 Jun 2025.
- **On Infinite Tensor Networks, Complementary Recovery and Type II Factors.** Talk, When Quantum Field Theory meets Quantum Information 2025 (UC3M, Madrid), 14–17 Dec 2025.